

Group 18 - Screen Standardization Requirements

1. Screen layout

a. General layout structure

- Header:
 - + Positioned at the top of the screen
 - + Contains the AIMS logo on the left
 - + Contains the screen title in the center
 - + Contains the menu bar or major actions on the right
- Navigation / utility area
 - + Located inside the header or directly below it.
 - + Includes functions such as Search, Filter, Sort, Cart, Back, and Account/Admin shortcuts
- Main content area: displays the main content of the screen
 - + Homepage: random product list / search results
 - + Product detail: detailed product information
 - + Cart screen: list of cart items
 - + Delivery form: shipping information form
 - + Invoice/payment screens: invoice details and QR/payment method
- Action area
 - + Located at the bottom of the content area or at the bottom of a form
 - + Contains major action buttons such as: Add to Cart, Update Cart, Place Order, Confirm, Cancel, Pay, Back
- Footer: Contains copyright information, support contact details, or system status

b. Minimum resolution and responsiveness

- Minimum desktop resolution: 1366 × 768 px
- Design width: 1440 px
- Responsive breakpoints:
 - + Desktop: ≥ 1200 px
 - + Mobile: < 768 px
- Responsive rules:
 - + The header must always preserve the logo, title, and primary menu/actions
 - + On mobile, the menu bar should be collapsed into a menu icon or action drawer
 - + Product grids should adapt from 4 columns → 2 columns → 1 column depending on screen width.
 - + Delivery forms and payment blocks should stack vertically on smaller screens.

- + Primary action buttons must remain visible and accessible within the viewport

c. Screen title

- Each screen must have one main title, placed at the center of the header
- The title must be concise and accurately reflect the screen function.
example: Home, Product Details, Your Cart, Delivery Information, Invoice, QR, Payment, Payment Result, Order Information.

d. Menu bar location

- The top-right corner is reserved for utility controls such as: Search, Filter, Sort, Cart, Back / Close
- For admin/manager screens, this area may contain: Product Management, Pending Orders, User Management, Logout

e. Standard button placement

- Primary action button: placed on the right side of the action area.
- Secondary action button: placed to the left of the primary button.
- Destructive actions such as Delete or Cancel Order must use a warning/danger color and be visually separated from the primary action.
- Back button:
 - + placed at the upper-left corner of the content area or within the action area
 - + must appear on all screens except the Home screen
- Close button (X):
 - + used only for popups/modals
 - + placed at the top-right corner of the popup

2. UI Kit

a. Font family

- Primary font: Roboto
- Fallback: Arial, sans-serif

b. Font size hierarchy

- Heading 1 / Main page title: 24 px, bold
- Heading 2 / Section title: 20 px, semibold
- Heading 3 / Card title / Popup title: 18 px, semibold
- Body text: 16 px, regular
- Secondary text / metadata: 14 px, regular
- Caption / helper text / validation hint: 12 px, regular
- Button text: 14–16 px, medium

c. Color palette

- Primary color: #0078D7
Used for primary buttons, main links, focus state, selected tabs
- Accent / highlight color: #FF6B35
Used for prominent CTAs such as Place Order, Pay Now, and important badges
- Success color: #2E7D32
Used for successful payment, successful order placement, and valid input
- Warning color: #ED6C02
Used for low stock warnings, pending issues, or caution messages
- Error color: #D32F2F
Used for validation errors, payment failures, and invalid operations
- Neutral colors
 - + Background: #F8F9FA
 - + Surface/Card: #FFFFFF
 - + Primary text: #212121
 - + Secondary text: #616161
 - + Border/Divider: #D9D9D9
- Color usage rules:
 - + Primary color is used for standard actions
 - + Accent color is used for high-emphasis actions
 - + Error/success/warning colors are used only for status communication

d. Buttons

- Primary button
 - + primary color background
 - + white text
 - + 8 px border radius
 - + minimum height: 40 px
- Secondary button
 - + white background
 - + primary color border
 - + primary color text
- Danger button
 - + red background
 - + white text
- Disabled button
 - + light gray background
 - + gray text
 - + non-clickable

e. Input fields

- Standard height: 40 px

- Border radius: 8 px
- Default border: 1 px solid neutral border
- Focus state: primary color border
- Error state: error color border + helper text below
- Placeholder uses secondary text color

f. Icons & graphics

- **Icon set:** Material Icons
- Common icons:
 - + search
 - + shopping_cart
 - + arrow_back
 - + close
 - + filter_list
 - + sort
 - + delete
 - + edit
 - + check_circle
 - + error
 - + info
- Rules:
 - + Important action icons should be accompanied by text labels
 - + Product images must keep the correct aspect ratio and image quality.
 - + If an image is missing or broken, a standard placeholder image must be displayed

g. Spacing and grid

- Standard spacing scale:
 - + 4 px
 - + 8 px
 - + 12 px
 - + 16 px
 - + 24 px
 - + 32 px
- Rules:
 - + Space between label and input: 8 px
 - + Space between form fields: 16 px
 - + Card padding: 16 px
 - + Space between major sections: 24–32 px

3. Data formatting

- **Numeric values:** Use comma separators every three digits (e.g. 1,000; 25,000)
- **Currency format:**
 - + Default display format: 1,000 VND
 - + In compact areas, 1,000^d may be used
 - + Only one format should be used consistently within the same screen
 - + If the value is zero, display 0 VND, not blank
 - + Clearly distinguish:
 - subtotal excluding VAT
 - total including VAT
 - delivery fee
 - final payable amount
- **Date format:** DD/MM/YYYY
- **Time and datetime format**
 - + Time format: HH:mm
 - + Datetime format: DD/MM/YYYY HH:mm
- **Phone number**
 - + Only numeric characters are allowed
 - + Length: 10–11 digits
 - + Letters and special characters are not allowed, except for display formatting
 - + example: 0987 654 321
- Email format: follow standard email format (example: customer@example.com)
- Product-related string handling
 - + Only valid UTF-8 characters may be displayed
 - + Dangerous characters must be removed or escaped to prevent rendering/UI injection issues
 - + Leading and trailing spaces must be trimmed
 - + Raw scripts/HTML must not be rendered directly
- Text truncation
 - + Product titles that are too long in card/list view:
 - maximum 2 lines
 - overflow should be shown with ellipsis . . .
 - + Full titles should be visible in a tooltip or on the detail screen

4. Control rules

a. Input validation process

The validation flow is standardized as follows:

- **Step 1: Empty check**
 - If a required field is left blank, an error must be shown immediately.

- **Step 2: Format check**

If the field is not empty, validate its format.

example: Email is invalid, Date format is invalid

- **Step 3: Business rule check**

After the input is valid in format, validate business rules.

example: "Quantity must be greater than 0", "Requested quantity exceeds available stock"

b. Validation display rules

- Errors must be displayed directly below the related field
- Invalid fields must change border color to red
- If the error applies to the whole form/screen, an error banner must also appear at the top of the form
- User-entered data must not be cleared after validation fails

c. Focus navigation sequence

- Tab order must follow a logical order from left to right, top to bottom.
- For the delivery form:
 1. Recipient name
 2. Phone number
 3. Email
 4. Province/City
 5. Address
 6. Delivery method
 7. Continue button
- The Back button should be focused after the primary form controls.
- When a popup opens, focus must be placed on the first control or the close button.

d. Navigation and screen transition rules

- **Standard application flow:**
 - + Opening screen (optional branding)
 - + Splash screen
 - + Home screen
- **Dependent child screen** (modal/popup) is used for:
 - + Product detail popup from homepage/product list
 - + Confirmation dialogs
 - + Help popup

When a popup is open, the parent screen must become non-interactive.
- **Simple transition** is used for:
 - + Cart → Delivery Form
 - + Delivery Form → Invoice

- + Invoice → QR Payment
- + QR Payment → Payment Result
- **Back navigation**
 - + must always be available in process screens
 - + should allow users to return to the previous step without losing unsaved information when appropriate
- **Close (X)**
 - + used only for popup/modal screens
 - + closes the popup and returns to the parent screen

e. Keyboard and navigation input

- No business shortcut keys are required
- The interface should support:
 - + Tab / Shift+Tab for moving focus
 - + Enter for submitting simple forms when appropriate
 - + Esc for closing popups if allowed
- Keyboard shortcuts must not directly trigger dangerous actions such as deleting items or making payment immediately

f. Rules for cart and order flow

- When the user clicks **Place Order**, the system must:
 - + check current stock availability
 - + if stock is insufficient, display a clear message showing the available quantity
 - + require the user to update the cart before continuing
- When the user changes the delivery address or delivery method:
 - + the system must automatically recalculate the delivery fee
 - + the invoice summary must be updated immediately
- Before moving to payment, a temporary invoice must be displayed including: item list, quantity, product prices, subtotal excluding VAT, total including VAT, delivery fee, final payable amount

g. Rules for payment screens

- The QR code must be displayed in the center of the payment area for easy scanning
- Supporting information must include: amount, transaction content/reference, order ID
- Buttons must include: Back, Switch Payment Method, Check Payment Status
- If payment fails or times out:
 - + a failure status must be displayed
 - + the user must be allowed to retry or choose another payment method

5. Other standardization

a. Error handling

- Error messages must be: short and clear, free of technical jargon where possible, and helpful in guiding the user to fix the issue

examples: "Phone number is required.", "Phone number format is invalid.", "The requested quantity exceeds available stock."

- The following vague messages must not be used: "Error occurred", "Invalid input", "System failed"

b. Confirmation rules

- Important actions must require user confirmation, such as:
 - Remove item from cart
 - Cancel order
 - Exit payment flow before completion
 - Delete product (manager side)
- Standard confirmation popup:
 - short title
 - message explaining the consequence of the action
 - two buttons: Confirm / Cancel

c. Loading and busy state:

For potentially time-consuming actions such as search, recalculating delivery fee, generating QR codes, and checking payment status

- a loading indicator should be displayed
- repeated submit actions should be disabled during loading
- the current context should remain visible
- duplicate requests caused by multiple clicks must be prevented

d. Empty states

- If no data is available, the system must show a clear empty state
example: "No products found.", "Your cart is empty.", "No pending orders."
- An empty state should include: an illustrative icon, a short descriptive sentence

e. Accessibility and readability

- Text/background contrast must remain readable
- Status must not be conveyed by color alone; icon/text should also be used
- Clickable button area must be at least 40 px in height
- Form labels must always be visible and not rely only on placeholders

f. Consistency in terminology

Terminology must remain consistent across all screens, mockups, popups, and specifications:

- Use “Cart” consistently, not sometimes “Basket”
- Use “Place Order” consistently
- Use “Delivery Information” consistently
- Clearly distinguish between “Invoice” and “Payment Transaction”
- Use “Pending Processing” consistently if showing order state to users/managers

g. Security-related UI rules

- Unnecessary sensitive information must not be displayed
- Raw technical exceptions from payment gateways must not be shown directly to users
- Transaction/reference content should display only the necessary part
- All user input must be sanitized before being rendered back on the UI